

NIKHIL DHILLON

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EDUCATION

University of Victoria

Bachelor of Science in Computer Science (Honours) | GPA: 8.14/9.0

Victoria, BC

Expected April 2027

Relevant Coursework: Data Structures & Algorithms, Communication & Networking, Database Systems, Data Mining

EXPERIENCE

University of Victoria

Incoming Teaching Assistant, Database Systems

Victoria, BC

September 2026 - December 2026

Reliable Controls

Software Developer Intern

Victoria, BC

January 2026 - August 2026

- Reduced MFC GDI resource usage by approximately 75% during a critical workflow, from 8,000+ objects to approximately 2,000, eliminating native Windows resource leaks
- Built a C++/MFC connection-cancellation dialog that let users immediately abort stalled connection attempts, eliminating 30–60 second timeout waits
- Built a CEF-powered HTML/SVG reporting system that generates high-resolution, print-ready visualizations of controller I/O
- Prototyped an XML-to-SQLite configuration migration using Entity Framework Core, including schema migrations and data synchronization

One Nation

Software Developer (Part-time)

Remote (Vancouver, BC)

April 2025 - July 2025

- Built a natural language-driven reporting pipeline using OpenAI API and PostgreSQL, enabling users to generate custom insights in under 20 seconds
- Implemented a Laravel Filament password-reset workflow with token validation, expiration handling, and secure credential updates
- Developed custom WordPress plugins to extend core CMS functionality, integrate with third-party APIs, and deliver performance-tuned solutions based on client-specific requirements

Reliable Controls

Software QA Analyst Intern

Victoria, BC

September 2024 - April 2025

- Designed and optimized test case workflows in TestRail, validating over 100 test cases for functional and non-functional requirements, and ensuring compliance with QA standards and testing best practices
- Streamlined API testing and integration validation workflows with Bruno, increasing reliability and adherence to API specifications, while reducing API validation errors by 20%
- Reduced QA review-cycle time by 30% by optimizing test workflows and introducing automation, improving sprint velocity

PROJECTS

DoNext | Next.js, TypeScript, FastAPI, PostgreSQL

[GitHub](#)

- Built an adaptive semester-planning platform that models courses, deadlines, commitments, goals, sleep, and availability to support dynamic semester scheduling
- Engineered a PDF/DOCX/TXT ingestion pipeline that extracts schedules and deadlines and converts course documents into structured planning data
- Developed a FastAPI/PostgreSQL backend with user-scoped APIs and persistent planning data, establishing the foundation for AI-assisted prioritization and dynamic replanning

GROWTH | React Native, Expo, TypeScript, Supabase

[Live](#) | [GitHub](#)

- Built and launched a live strength analytics platform with 7 active users, enabling workout logging, PR tracking, and long-term strength progression analysis
- Developed a custom performance-scoring engine combining estimated 1RM, training volume, and fatigue resistance
- Implemented Supabase authentication and persistent workout data, alongside friend leaderboards, exercise-specific PR tracking, and progress visualizations

TECHNICAL SKILLS

- **Programming Languages:** C++, C#, Python, Java, JavaScript/TypeScript, SQL, C, PHP, HTML/CSS
- **Frameworks and Libraries:** MFC, .NET, React, React Native, Next.js, Node.js, Express.js, FastAPI, Django, PyTorch, Entity Framework Core
- **DevOps and Tools:** Git, Docker, Linux, CI/CD, Microsoft Azure, Google Cloud Platform, AWS (EC2, S3, Lambda, RDS)
- **Databases and Backend:** PostgreSQL, SQLite, Supabase
- **Project Management and Collaboration:** Jira, Confluence, Scrum, Agile